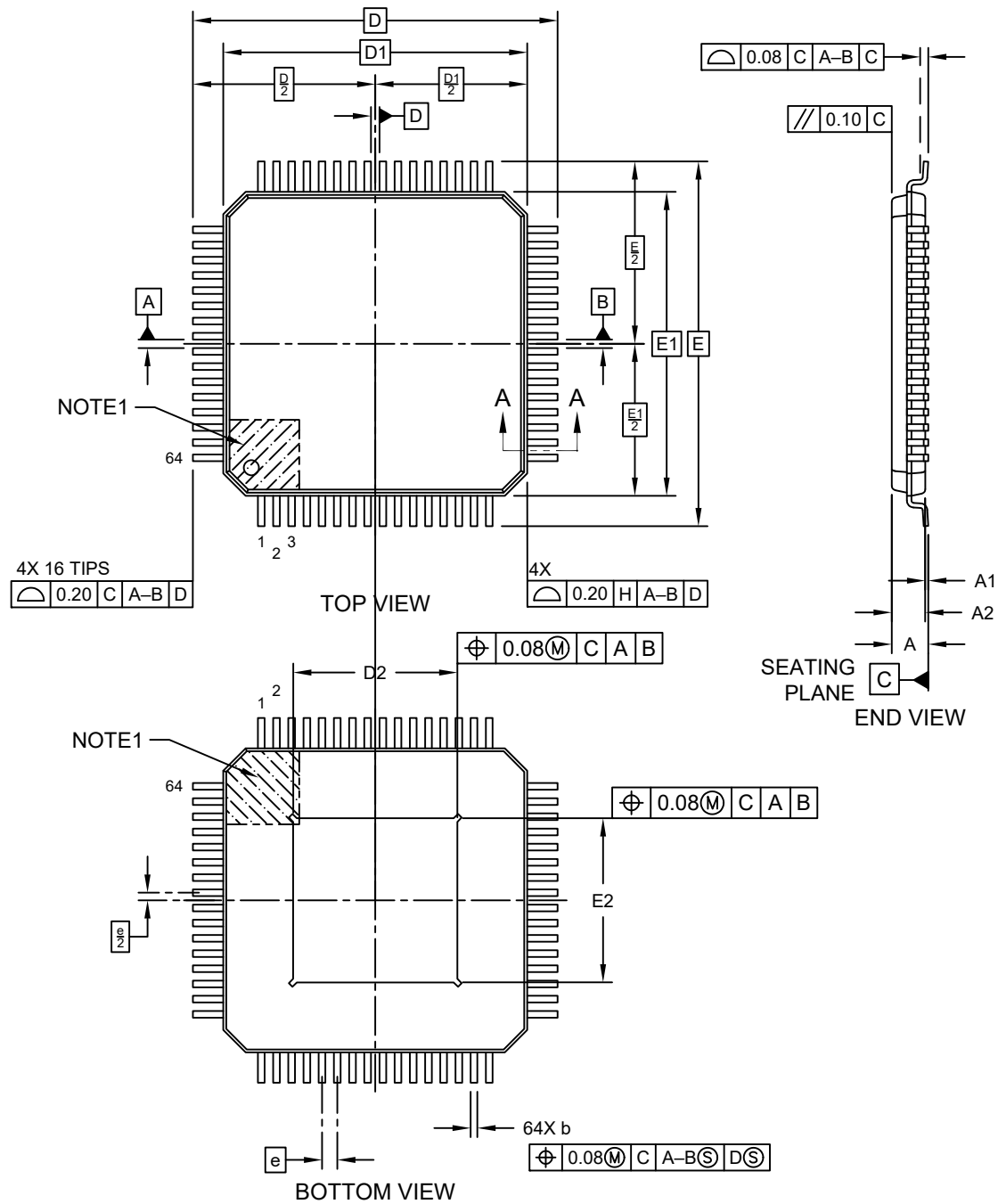


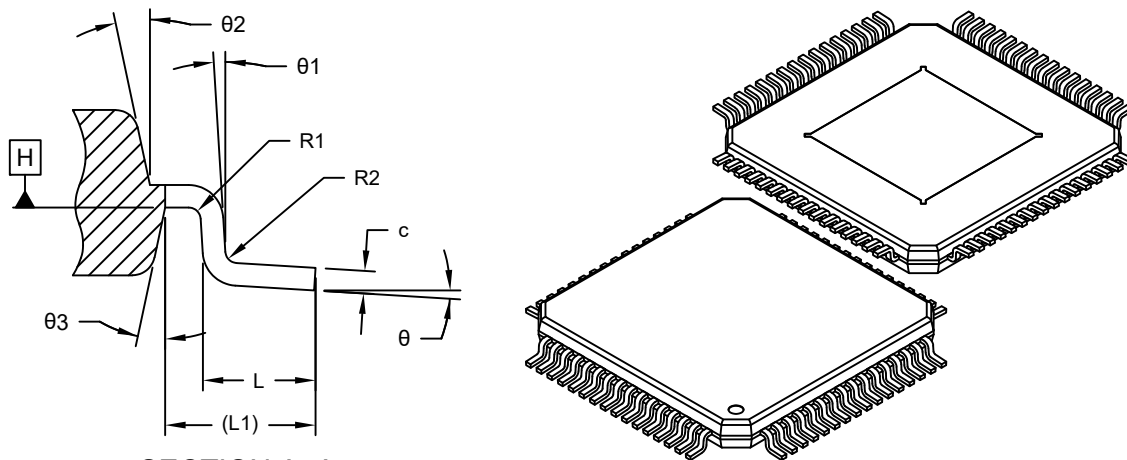
# **64-Lead Plastic Thin Quad Flat Pack (XDB) - 10x10x1.0 mm Body [TQFP]** **With 5.4 mm Exposed Pad**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 64-Lead Plastic Thin Quad Flat Pack (XDB) - 10x10x1.0 mm Body [TQFP] With 5.4 mm Exposed Pad

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**SECTION A-A**

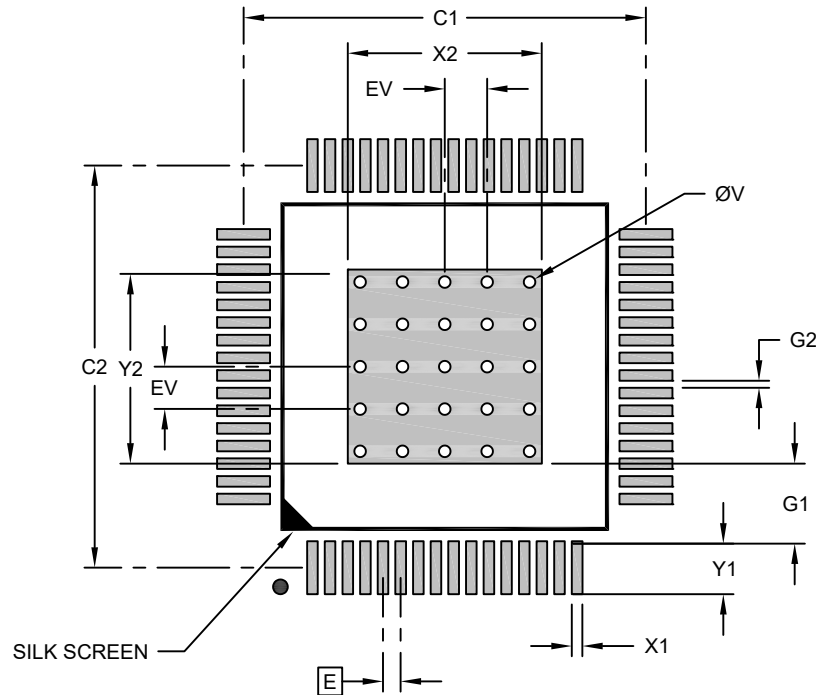
| Dimension Limits         | Units | MILLIMETERS |      |      |
|--------------------------|-------|-------------|------|------|
|                          |       | MIN         | NOM  | MAX  |
| Number of Terminals      | N     | 64          |      |      |
| Pitch                    | e     | 0.50 BSC    |      |      |
| Overall Height           | A     | -           | -    | 1.20 |
| Standoff                 | A1    | 0.05        | -    | 0.15 |
| Molded Package Thickness | A2    | 0.95        | 1.00 | 1.05 |
| Overall Length           | D     | 12.00 BSC   |      |      |
| Molded Package Length    | D1    | 10.00 BSC   |      |      |
| Exposed Pad Length       | D2    | 5.30        | 5.40 | 5.50 |
| Overall Width            | E     | 12.00 BSC   |      |      |
| Molded Package Width     | E1    | 10.00 BSC   |      |      |
| Exposed Pad Width        | E2    | 5.30        | 5.40 | 5.50 |
| Terminal Width           | b     | 0.17        | 0.22 | 0.27 |
| Terminal Thickness       | c     | 0.09        | -    | 0.20 |
| Terminal Length          | L     | 0.45        | 0.60 | 0.75 |
| Footprint                | L1    | 1.00 REF    |      |      |
| Lead Bend Radius         | R1    | 0.08        | -    | -    |
| Lead Bend Radius         | R2    | 0.08        | -    | 0.20 |
| Foot Angle               | θ     | 0°          | 3.5° | 7°   |
| Lead Angle               | θ1    | 0°          | -    | -    |
| Mold Draft Angle         | θ2    | 11°         | 12°  | 13°  |
| Mold Draft Angle         | θ3    | 11°         | 12°  | 13°  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 64-Lead Plastic Thin Quad Flat Pack (XDB) - 10x10x1.0 mm Body [TQFP] With 5.4 mm Exposed Pad

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Dimension                        | Units | MILLIMETERS |       |      |
|----------------------------------|-------|-------------|-------|------|
|                                  |       | MIN         | NOM   | MAX  |
| Contact Pitch                    | E     | 0.50 BSC    |       |      |
| Center Pad Width                 | X2    |             |       | 5.50 |
| Center Pad Length                | Y2    |             |       | 5.50 |
| Contact Pad Spacing              | C1    |             | 11.40 |      |
| Contact Pad Spacing              | C2    |             | 11.40 |      |
| Contact Pad Width (Xnn)          | X1    |             |       | 0.30 |
| Contact Pad Length (Xnn)         | Y1    |             |       | 1.50 |
| Contact Pad to Center Pad (Xnn)  | G1    | 2.20        |       |      |
| Contact Pad to Contact Pad (Xnn) | G2    | 0.20        |       |      |
| Thermal Via Diameter             | V     |             | 0.33  |      |
| Thermal Via Pitch                | EV    |             | 1.20  |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23558 Rev A